

L13b: Interpretable Allocation and Model-Predictive Rebalancing

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Objectives for Today

Today, we turn the utility allocation of L8b into a book that trades every day. How do we build such a book, and how do we judge it when the real market gives us only one path?

Today's concepts

- What to hold: a score from the single index model and the market state, and a picker.
- How to weight and when to trade: maximum utility under limits, and a controller that re-plans when the book leaves its forecast.
- How to judge it: one real path against many synthetic markets.

What You Need to Know

Core concepts

- The score $\gamma_i(t)$ has the sign of the stock's expected growth rate; the market state ξ_t only changes its size.
- A re-plan maximizes a utility of the share counts under a budget and limits on weight, volatility, and turnover. The book then holds until the 21-day horizon, a band exit, or an 8% drawdown.
- A backtest is one path; synthetic markets and sell-off sets give many.

For understanding (you won't reproduce these)

- The online update, the forecast covariance, the cost model, the hedge pool, and the fit of the generator's β drift.

Lectures and Examples

Lecture:

CHEME-5660-L13b-Lecture-MPC-Allocation-Fall-2026.ipynb

Examples:

CHEME-5660-L13b-Example-MPC-RealData-Fall-2026.ipynb

CHEME-5660-L13b-Example-MPC-SyntheticData-Fall-2026.ipynb

From a Utility Allocation to a Live Book

In L8b, we chose share counts once, by maximizing a utility under a budget. A book that trades for a year and a half has three more problems:

- What to hold: hundreds of stocks, and estimates that change daily.
- How to weight: under the budget alone, a utility can concentrate the book or take on more volatility than the market.
- When to trade: trading daily costs money; a fixed calendar ignores the market in between.

The INFORMS 2026 study (arXiv:2611.03022) answers each in a way we can inspect. It calibrates on 2014–2024 and trades 373 stocks from January 2025 to September 2026.

What to Hold: The Market Inputs

Each stock's single index model, $g_i = \alpha_i + \beta_i\, g_M + \varepsilon_i$, is updated every day, giving $\alpha_i(t)$ (1/yr) and $\beta_i(t)$. From the `SPY` close, as in L8b, the recent market growth $\tilde{g}_{M,t}$ (1/yr) is its 252-day moving average, and the market state is:

$$\xi_t = -G\left(\frac{\bar{S}_t^{(21)}}{\bar{S}_t^{(63)}} - 1\right)$$

where $\bar{S}_t^{(21)}$ and $\bar{S}_t^{(63)}$ are 21-day and 63-day moving averages of the close, and the gain G makes $|\xi_t| \leq 1$ on 99% of 2014–2024 days. A falling market has $\xi_t > 0$.

The γ Score

The score of stock i on day t is:

$$\gamma_i(t) = \tanh\left(\frac{\tilde{g}_{i,t}}{\max(|\beta_i(t)|,\, 0.1)^{\xi_t}}\right), \qquad \underbrace{\tilde{g}_{i,t} = \alpha_i(t) + \beta_i(t)\,\tilde{g}_{M,t}}_{\text{expected growth rate}}$$

- Sign: the denominator is positive, so $\gamma_i(t)$ has the sign of $\tilde{g}_{i,t}$.
- Size: when $\xi_t > 0$, the score favors stocks with $|\beta_i| < 1$; when $\xi_t < 0$, stocks with $|\beta_i| > 1$.

L8b assumed $\beta_i > 0$; the floor of 0.1 keeps the score finite for any β_i.

The Pickers

A picker chooses the names $\mathcal{N}$ the book may hold. In our examples:

- γ per sector: the 3 stocks with the highest $\gamma_i(t)$ in each of the 11 sectors, 33 names.
- Random per sector: 3 stocks drawn at random in each sector, a baseline with the same sector balance.
- A fixed list: names chosen in advance, such as the 13 stocks of L8b.

The picker runs on the first day, every 21 trading days after its last pick, and whenever the book's value leaves its forecast band.

How to Weight: Maximum Utility Under Limits

With shares n_i, close S_i, wealth W, and weight $w_i = n_i S_i / W$, a re-plan solves:

$$\max_{n \geq 0} \; U(n;\gamma) \quad \text{s.t.} \quad \underbrace{\textstyle\sum_i n_i S_i \leq W}_{\text{budget}}, \qquad \underbrace{w_i \leq \bar{w}_i}_{\text{cap}},$$

$$\underbrace{\sqrt{w^\top \Sigma_g w} \leq \sigma_{\text{SPY}}}_{\text{volatility}}, \qquad \underbrace{\bar{c}\,\bar{S} \textstyle\sum_i |n_i - n_i^{\text{prev}}| \leq B}_{\text{turnover}}$$

where $\bar{w}_i = \max(0.10, 1.5/K)$ for K picked names, Σ_g is the covariance of the daily growth rates, σ_{SPY} the volatility of `SPY` on 2014–2024, n^{prev} the shares held, $\bar{c} = 5 \times 10^{-4}$ the half spread, $\bar{S}$ the mean price, and $B = 0.10\,W$. Uninvested wealth is cash.

The Weighting Rules

Four utilities of L8b's kind use the names with a positive score, $\mathcal{A}^+$:

- Cobb–Douglas: $U = \prod_{i \in \mathcal{A}^+} n_i^{\gamma_i}$, the L8b allocation.
- CES on share counts ($\eta = 0.5$) and CES on weights.
- Stone–Geary: Cobb–Douglas above a 1% baseline in every name.

Three standard portfolios compare: minimum variance, the tangency portfolio, and a 130/30 long–short book.

With the limits there is no closed form, but each problem is a convex cone program, solved with Clarabel (SCS as the fallback). If both fail, the book keeps its holdings.

When to Trade: The Forecast Band

At a re-plan, the controller forecasts the wealth of holding the targets $n^\star$ with a multi-asset geometric Brownian motion in growth rates: mean $\tilde{g}_{i,t}$, covariance Σ_g, $N_p = 1{,}000$ paths, $T = 21$ trading days. With $\mu_W(\tau)$ and $\sigma_W(\tau)$ the mean and SD of the forecast wealth τ days later, the band is:

$$\mu_W(\tau) - z\,\sigma_W(\tau) \;\le\; W(t) \;\le\; \mu_W(\tau) + z\,\sigma_W(\tau), \qquad z = 1.96$$

If the model were right, the wealth would stay inside on about 95% of days.

The Triggers

After each close, with τ the days since the last re-plan and W^{pk} the running peak, the controller re-plans on the first of:

1. Drawdown: $1 - W(t)/W^{\text{pk}} > D_{\max} = 0.08$; the peak then resets.
2. Horizon: $\tau \geq T$, the plan has run its 21 days.
3. Band exit: the wealth has left the forecast band.

A scheduled pick that changes the names also re-plans. Between re-plans the book does not trade: the band is a no-trade region on its value, and each re-plan starts a new forecast from the measured wealth.

Trades and Costs

Orders fill at the next open, sells before buys, in fractional shares. A fill of q shares of name i at the open price S_i^o costs:

$$C(q) = \underbrace{\bar{c}\, S_i^o\, |q|}_{\text{half spread}} + \underbrace{\kappa \frac{|q|}{A_i} S_i^o\, |q|}_{\text{market impact}}$$

where $\bar{c} = 5 \times 10^{-4}$, A_i is the stock's mean daily volume over 63 trading days (shares/day), and $\kappa = 0.001$. The impact grows with the share of the day's volume that the order takes.

The Market Hedge

With the hedge on, every re-plan also limits the book's market loading:

$$\sum_{i \in \mathcal{N}} \beta_i \, w_i \;\leq\; \bar{\beta} = \max\left(0,\, 1 - \xi_t\right)$$

The cap is the loading of `SPY` at $\xi_t = 0$, tightens as the market falls, and reaches zero at $\xi_t \geq 1$. To meet it, the book can hold cash or names from a hedge pool: up to five stocks whose online β_i and down-day β over the last year are at most zero. The hedge trades growth in rising markets for protection in falling ones.

How to Judge the Method

On real data, the method runs once, on one path with one broad sell-off in spring 2025. A choice that did well may have been lucky.

The study also uses the L13a generator, fitted to the same 373 stocks on 2014–2024, with two parts tied to the method:

- Drifting β: sized so the method's online β moves as much on synthetic stocks as on real ones.
- Intercepts: each α_i pulled toward the mean, so they spread and persist as the real ones do.

Each market calibrates on its 2014–2024 and trades its 2025–2026.

Typical and Sell-off Sets

From 10,000 candidate markets, the study chooses three sets by the `SPY` path over the trading window:

- Typical: 600 markets drawn at random.
- Burst sell-offs: the 200 deepest drawdowns among markets with a burst of extreme daily moves.
- Bear sell-offs: the 200 deepest drawdowns among markets without one, with slower, more sustained declines.

The results are read as distributions, set by set, against `SPY` and an equal-weight portfolio of every stock on the same market.

What the Study Found

- Stocks over weights: on real prices, one 33-stock selection ended between 1.09 and 1.23 times initial wealth across the seven rules; different selections ended between 1.03 and 1.70.
- Early re-plans: they made the median worst drawdown smaller in most combinations, raised trading cost, and did not consistently raise wealth.
- Sell-offs: every combination, random picks included, ended above simulated `SPY` and the equal-weight universe in the median.
- Caveats: the generator's persistent trends favored the γ picker more than the real market did; the hedge cost wealth on real prices and added to it in the bear set.

Summary

We built a book that chooses, weights, and trades stocks every day, and tested it on real and synthetic markets.

Key takeaways

- What to hold: a bounded score from the single index model and the market state, and a picker.
- How to weight and when to trade: utility under limits as a convex program, and re-plans at the horizon, a band exit, or a drawdown.
- How to judge: one real path, then distributions over typical and sell-off synthetic markets.

The method joins the single index model, utility allocation, and synthetic markets.